

```

(kali㉿kali)-[~]
$ mkdir -p ~/CIP-B102-Lab7-Steganography

(kali㉿kali)-[~]
$ cd ~/CIP-B102-Lab7-Steganography

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ wget -O tower_original_image_for_lab.bmp "https://drive.google.com/uc?export=download&id=1CE-4EOe9-HEF_9mg3Y4N7u15qUbILDm0"
--2026-07-15 12:35:36-- https://drive.google.com/uc?export=download&id=1CE-4EOe9-HEF_9mg3Y4N7u15qUbILDm0
Resolving drive.google.com (drive.google.com) ... 142.250.129.100, 142.250.129.101, 142.250.129.102, ...
Connecting to drive.google.com (drive.google.com)|142.250.129.100|:443 ... connected.
HTTP request sent, awaiting response ... 303 See Other
Location: https://drive.usercontent.google.com/download?id=1CE-4EOe9-HEF_9mg3Y4N7u15qUbILDm0&export=download [following]
--2026-07-15 12:35:37-- https://drive.usercontent.google.com/download?id=1CE-4EOe9-HEF_9mg3Y4N7u15qUbILDm0&export=download
Resolving drive.usercontent.google.com (drive.usercontent.google.com) ... 142.251.39.193, 2a00:1450:4006:814::2001
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)|142.251.39.193|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 9277494 (8.8M) [application/octet-stream]
Saving to: 'tower_original_image_for_lab.bmp'

tower_original_image 100%[=====>] 8.85M 1.15MB/s in 8.6s

2026-07-15 12:35:51 (1.03 MB/s) - 'tower_original_image_for_lab.bmp' saved [9277494/9277494]

```

```

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ ls -lh
total 8.9M
-rw-rw-r-- 1 kali kali 8.9M Dec  5 2025 tower_original_image_for_lab.bmp

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ file tower_original_image_for_lab.bmp
sha256sum tower_original_image_for_lab.bmp > lab7_hashes.txt
tower_original_image_for_lab.bmp: PC bitmap, Windows 3.x format, 1365 x 2265 x 24, resolution 2834 x 2834 px/m, cbSize 9277494, bits offset 54

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat lab7_hashes.txt
6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db tower_original_image_for_lab.bmp

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$

```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat > secret.txt << 'EOF'
ICDFA CIP-B102 Lab 7 Evidence Note
Student Name: Fuseini Imoru
Message: This file was hidden and recovered during a controlled steganography lab.
EOF

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat secret.txt
ICDFA CIP-B102 Lab 7 Evidence Note
Student Name: Fuseini Imoru
Message: This file was hidden and recovered during a controlled steganography lab.

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ file secret.txt
secret.txt: ASCII text
```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ sha256sum secret.txt >> lab7_hashes.txt

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat lab7_hashes.txt
6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db  tower_original_image_for_lab.bmp
b4b03641c0b06c09cc1a1274476f075e8bfdfa7e2650f9ec6e8c24d75bbc41b8  secret.txt

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$
```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ steghide embed \
  -ef secret.txt \
  -cf tower_original_image_for_lab.bmp \
  -sf tower_stego_lab7.bmp \
  -p 1234
embedding "secret.txt" in "tower_original_image_for_lab.bmp" ... done
writing stego file "tower_stego_lab7.bmp" ... done

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ ls -lh tower_original_image_for_lab.bmp tower_stego_lab7.bmp
-rw-rw-r-- 1 kali kali 8.9M Dec  5  2025 tower_original_image_for_lab.bmp
-rw-rw-r-- 1 kali kali 8.9M Jul 15 12:43 tower_stego_lab7.bmp
```

```

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ file tower_stego_lab7.bmp
tower_stego_lab7.bmp: PC bitmap, Windows 3.x format, 1365 x 2265 x 24, resolution 28
34 x 2834 px/m, cbSize 9277494, bits offset 54

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ sha256sum tower_stego_lab7.bmp >> lab7_hashes.txt

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat lab7_hashes.txt
6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db  tower_original_ima
ge_for_lab.bmp
b4b03641c0b06c09cc1a1274476f075e8bfdfa7e2650f9ec6e8c24d75bbc41b8  secret.txt
c7669807d6bce5ceb52c3045e6149b40d89002ba903228c66092cbb8a2e55d23  tower_stego_lab7.b
mp

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$

```

```

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ mkdir -p extracted

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ steghide extract \
-sf tower_stego_lab7.bmp \
-xf extracted/extracted_secret.txt \
-p 1234
wrote extracted data to "extracted/extracted_secret.txt".

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat extracted/extracted_secret.txt
ICDFA CIP-B102 Lab 7 Evidence Note
Student Name: Fuseini Imoru
Message: This file was hidden and recovered during a controlled steganography lab.

```

```

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ sha256sum extracted/extracted_secret.txt >> lab7_hashes.txt

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ diff secret.txt extracted/extracted_secret.txt && echo "MATCH: extracted file is
identical"

MATCH: extracted file is identical

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ sha256sum secret.txt extracted/extracted_secret.txt
b4b03641c0b06c09cc1a1274476f075e8bfdfa7e2650f9ec6e8c24d75bbc41b8  secret.txt
b4b03641c0b06c09cc1a1274476f075e8bfdfa7e2650f9ec6e8c24d75bbc41b8  extracted/extrac
ted_secret.txt

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$

```



```

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ ls -lh tower_original_image_for_lab.bmp tower_stego_lab7.bmp
-rw-rw-r-- 1 kali kali 8.9M Dec  5 2025 tower_original_image_for_lab.bmp
-rw-rw-r-- 1 kali kali 8.9M Jul 15 12:43 tower_stego_lab7.bmp

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ sha256sum tower_original_image_for_lab.bmp tower_stego_lab7.bmp
6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db tower_original_image_for_lab.bmp
c7669807d6bce5ceb52c3045e6149b40d89002ba903228c66092cbb8a2e55d23 tower_stego_lab7.bmp

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ xxd -l 64 tower_original_image_for_lab.bmp
00000000: 424d 3690 8d00 0000 0000 3600 0000 2800  BM6 .. ....6 ... (.
00000010: 0000 5505 0000 d908 0000 0100 1800 0000  ..U. .. .. ...
00000020: 0000 0000 0000 120b 0000 120b 0000 0000  .... .. .. ...
00000030: 0000 0000 0000 1e22 3520 2338 1e20 381f  ......"5 #8. 8.

```

```

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ xxd -l 64 tower_stego_lab7.bmp
00000000: 424d 3690 8d00 0000 0000 3600 0000 2800  BM6 .. ....6 ... (.
00000010: 0000 5505 0000 d908 0000 0100 1800 0000  ..U. .. .. ...
00000020: 0000 0000 0000 120b 0000 120b 0000 0000  .... .. .. ...
00000030: 0000 0000 0000 1e22 3520 2338 1e20 381f  ......"5 #8. 8.

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ strings tower_stego_lab7.bmp | head -n 20
"5 #8
:;%?()E')H*+M*,N).O(.Q'0R(1S(2T/9[5Ac=NoDYyF_
B_~;Wv7Nn2Cd2>`1:\17Z79\79\;;_;;_;;='79\67]56\56\67]67]56\56\56\79\57Z46Y57Z46Y68[68[
35X56X23U23U45W23U12T34V34V56X45W34V-.P./Q,-O/0R/0R32R10P/.N.-M/.N/.N..L--K//M++I..L
--K,-I,-I*+G*+G),H8)E8)E'*F8*C'+D)*D')A(*B8(@')A')A8)>8)>*>$'<+,A+,A*+@(>)'(=((>
)=(');%`9#%7#%7#%7"$6$888)8$'6%%5%%5##3##3$4!!1!/
#!#+-/??BGMPNTNPQJLLFHHHJJIKKJLLLNIIKKBD?AA;:<?>@EDF?>@=<?>@;:≤578:≤6899;<9;
<@ABDEJNOKOPMOPLNOIJNJKOORVRUYRWZ^adknlsvrx}msxZ^cJNS/47$),

```

```

'0> )78/<5>K3=G*4>5?IBLVEOYIS]OYcPZdP\hQ]iP\hMYeFR^<HT<HTEQ]IUaLXd0[gNZfJVbGS_FR^GS_
GS_7CO;/G4@L6BN-9E'3?(4@*6B.:F3?K:FRAMYDP\EQ]DP\FPZEYOYEN[EN[FO\HQ^HQ^DMZBLVDNXGQ[GQ[
FPZHR\IR_IR_HQ_GP^FO]EN\DM[DM[EN\GP^KtBIR_FO\FO\GP]FO\EN[GP]FO\DMZCLYEN[IR_KtAIR_HQ^
JS`JS`KtAKtALubPYfR[h0XeMV`PYcU^hYbLZcm\eo]fp\eo[dn\eo_hrbkudmwhq{ir|gpzjs}kt~lt

```

```

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ steghide info tower_stego_lab7.bmp
"tower_stego_lab7.bmp":
  format: Windows 3.x bitmap
  capacity: 377.4 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
  embedded file "secret.txt":
    size: 146.0 Byte
    encrypted: rijndael-128, cbc
    compressed: yes

```

```

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$

```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat > lab7_wordlist.txt << 'EOF'
password
admin
icdfa
cybersecurity
1234
letmein
EOF
```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat lab7_wordlist.txt
password
admin
icdfa
cybersecurity
1234
letmein
```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ stegseek tower_stego_lab7.bmp lab7_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "1234"
[i] Original filename: "secret.txt".
[i] Extracting to "tower_stego_lab7.bmp.out".
```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ ls -lh
total 18M
drwxrwxr-x 2 kali kali 4.0K Jul 15 12:46 extracted
-rw-rw-r-- 1 kali kali 360 Jul 15 12:46 lab7_hashes.txt
-rw-rw-r-- 1 kali kali 48 Jul 15 12:55 lab7_wordlist.txt
-rw-rw-r-- 1 kali kali 146 Jul 15 12:39 secret.txt
-rw-rw-r-- 1 kali kali 8.9M Dec 5 2025 tower_original_image_for_lab.bmp
-rw-rw-r-- 1 kali kali 8.9M Jul 15 12:43 tower_stego_lab7.bmp
-rw-rw-r-- 1 kali kali 146 Jul 15 12:56 tower_stego_lab7.bmp.out
```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
```

```
$ cat > fuseini_imoru_hidden_note.txt << 'EOF'
```

Steganography is the practice of hiding information inside another file, such as an image, audio file, or video file.

In digital forensics, steganography matters because investigators may encounter files that appear normal but secretly contain hidden evidence, confidential information, or malicious content. Understanding how to detect and recover hidden data is an important forensic skill.

EOF

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
```

```
$ cat fuseini_imoru_hidden_note.txt
```

Steganography is the practice of hiding information inside another file, such as an image, audio file, or video file.

In digital forensics, steganography matters because investigators may encounter files that appear normal but secretly contain hidden evidence, confidential information, or malicious content. Understanding how to detect and recover hidden data is an imp

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
```

```
$ steghide embed \  
-ef fuseini_imoru_hidden_note.txt \  
-cf tower_original_image_for_lab.bmp \  
-sf tower_stego_partG.bmp \  
-p 1234
```

embedding "fuseini_imoru_hidden_note.txt" in "tower_original_image_for_lab.bmp" ... 0
embedding "fuseini_imoru_hidden_note.txt" in "tower_original_image_for_lab.bmp" ... done
writing stego file "tower_stego_partG.bmp" ... done

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
```

```
$ sha256sum fuseini_imoru_hidden_note.txt
```

d51d6b45d8504a48de9144bd885adaae98315ace6ffd862195593b4b14b3cfa5 fuseini_imoru_hidden_note.txt

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
```

```
$ sha256sum tower_stego_partG.bmp
```

3c9def718a7d0e58dbe27cf7b192224354d6261cd756f601316454bf4491b57d tower_stego_partG.bmp

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
```

```
$ mkdir -p partG_extract
```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
```

```
$ steghide extract \  
-sf tower_stego_partG.bmp \  
-xf partG_extract/recovered_note.txt \  
-p 1234
```

wrote extracted data to "partG_extract/recovered_note.txt".

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ sha256sum fuseini_imoru_hidden_note.txt
d51d6b45d8504a48de9144bd885adaae98315ace6ffd862195593b4b14b3cfa5  fuseini_imoru_hidd
en_note.txt

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ sha256sum partG_extract/recovered_note.txt
d51d6b45d8504a48de9144bd885adaae98315ace6ffd862195593b4b14b3cfa5  partG_extract/reco
vered_note.txt

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ diff fuseini_imoru_hidden_note.txt partG_extract/recovered_note.txt \
&& echo "MATCH: files are identical"
MATCH: files are identical
```

```
(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ cat > partG_wordlist.txt << 'EOF'
password
admin
forensics
cybersecurity
1234
letmein
EOF

(kali㉿kali)-[~/CIP-B102-Lab7-Steganography]
$ stegseek tower_stego_partG.bmp partG_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "1234"
[i] Original filename: "fuseini_imoru_hidden_note.txt".
[i] Extracting to "tower_stego_partG.bmp.out".
```